

MD FARDIN AHMED

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ACADEMIC PROFILE

Electrical and Electronic Engineering graduate from BUET and MSc researcher at INRS EMT, Université du Québec, Canada. Experimental research experience spans quantum photonics, nanofabrication, and semiconductor characterization, complemented by VLSI industry experience in RTL to GDSII flow, TSMC 16 nm FinFET technology, Analog design in Cadence. Author of 10 peer reviewed publications in quantum photonics, solid state defect physics, and nanofabrication from 2025 to 2026. Recipient of the NSERC CGS M Scholarship for 2026, one of Canada's most competitive national graduate awards, in addition to full MSc funding. Brings teaching, mentoring, and academic outreach experience, with strong communication, student support, and research skills.

EDUCATION

Université du Québec - National Institute of Scientific Research (INRS) Jan 2025 – Present
M.Sc. in Energy, Materials, and Telecommunications (EMT)

Thesis writing in progress; coursework, research requirements, defense completed.

Bangladesh University of Engineering and Technology (BUET) Apr 2019 – Jul 2024

B.Sc. in Electrical and Electronic Engineering, **Electronics Major**

GPA: 3.73/4.00 (Last two terms: 3.83, 3.75)

RESEARCH EXPERIENCE

Research Assistant, INRS, Université du Québec - Experimental Research Jan 2025 – Present

- Led the end to end development of a **quantum optics platform**, building **confocal microscopy** and **HBT interferometry** systems for single photon characterization in **quantum photonics**. Also developed python based automation and perturbation compensation that enabled **99.8% single photon purity at room temperature**, addressing a major challenge for **quantum communication and sensing**. Presented in **USA and Canada**, received **national awards**. Currently preparing **US patent applications** and manuscripts for **high impact publications**, connecting material growth with real world quantum applications.
- Additional research work included different experimental studies on **GaN nanowire and semiconductor optoelectronic devices**, focusing on enhanced photon extraction from point defect single photon emitters, carrier dynamics in **InGaN p-i-n photodiodes**, and high responsivity **Co₃O₄ modified GaN nanowire** photodetectors.

Undergraduate Researcher, BUET - Simulation Based Research Aug 2023 – July 2024

- Designed and inversely optimized a **THz metamaterial** sensor for detecting trace pesticides using **FDTD** and **DFT**, achieving 99.43% absorption at 1.844 THz. Applied optimization algorithms including **Trust Region Framework** and **Particle Swarm** to refine the structure.
- Developed and verified RTL code for SPI design, designed a high-performance 1:8 Analog De-multiplexer, and performed layout design and verification for **Bandgap Reference, PLL, and High-Speed Digital IO RX/TX** circuits using **Cadence** tools.
- Applied **advanced ML/DL** and numerical optimization techniques in projects and **Kaggle competitions** spanning **material synthesis, GHI forecasting, and AI-based traffic control**. Implemented models in **Jupyter** using **TensorFlow, PyTorch, and scikit-learn**..[GitHub]

INDUSTRY EXPERIENCE

IC Layout Design Training, Ulkasemi, Canada March, 2026

Completed industry focused training in **full custom IC layout design** using **Cadence Layout XL** on **TSMC 16 nm FinFET (N16FF+)** technology, used in production of **Apple A10 Fusion SoC**. Designed and verified digital standard cell layouts and performed **DRC** and **LVS signoff**.

Industrial Attachment, QUEST Global, Bangladesh November, 2023

Gained hands-on experience in the VLSI design flow, using EDA tools (Cadence, Synopsys) for RTL design, synthesis, physical design, and DFT, while optimizing power-performance-area (PPA) trade-offs.

AWARDS AND RECOGNITION

- **SPIE Teddi Laurin Scholarship** May 2026
Awarded **USD 5000** by SPIE and Photonics Media, an international award recognizing outstanding research potential and leadership in optics and photonics.
- **NSERC CGS-M Scholarship** Apr 2026
Awarded **CAD 27000**, Canada's most prestigious national merit award for graduate research.
- **CMC Microsystems MNT MicroFAB Fabrication Award** Jun 2025
Awarded **CAD 5,000** to support advanced microfabrication research.
- **Best Presentation Award, Master's Thesis Seminar** Apr 2026
Received **CAD 500** in recognition of outstanding thesis presentation to the broad scientific community.
- **RISE Research Grant, BUET** Dec 2023
Awarded **BDT 46,920** for research on *Terahertz Metamaterials for Trace Pesticide Detection*.
- **Champion, Poster Presentation Competition, Department of EEE, BUET** Feb 2024
Won **BDT 5000** for clearly and effectively presenting research to a broad audience
- **Top Finalist, IEEE PES BDC Datathon** Jul 2020
Developed an advanced ML model for GHI prediction, from Helioscale omega station solar data (Feni).
- **Dean's List Merit Stipend, BUET** Apr 2020
Awarded for academic performance.

PUBLICATIONS

- **Md Fardin Ahmed**, N. Anand, MZ Baten, I. Aharonovich and SM Sadaf, "**On the photophysics and room temperature quantum emission of various epitaxial p-type GaN,**" SPIE Photonics West 2026, San Francisco, CA, USA. [link]
- **Md Fardin Ahmed**, MZ Baten, and SM Sadaf, "**Optical Emission Dynamics of Room Temperature GaN Quantum Defects: Pathways to Brighter and On Demand Single-Photon Emission,**" Electronic Materials Conference 2026, University of Michigan, Ann Arbor, Michigan, USA (**Accepted**).
- **Md Fardin Ahmed**; Anand, N; Jenson, C; Baten, M.Z.; and Sadaf, S.M. "**Deterministic III-Nitride Single-Photon Emitters in Nanowires for Quantum Technologies,**" - Forum québécois sur les matériaux avancés (FQMA) 2025, Québec, Canada. [link]
- **Md Fardin Ahmed**, MA Azim, and Ahmed Zubair, "**High sensitivity terahertz metamaterial sensor for trace pesticide detection,**" RSC Advances (2025). [link]
- SM Sadaf, N. Anand, **Md Fardin Ahmed**, DK Ghosh, E. Carbone, and H. Tang, "**On Apparent Absence of Green Gap in InGaN/GaN Quantum Disks and Wells Grown by Plasma-Assisted Molecular Beam Epitaxy,**" Journal of Nanophotonics. [link]
- A Ghosh, N Anand, **Md Fardin Ahmed**, and SM Sadaf, "**Engineering the Surface Recombination Velocity in Nanoscale InGaN p-i-n Photodiodes,**" Photonics North 2026, Quebec, Canada (**Accepted**).
- AFM Rasul, **Md Fardin Ahmed**, DK Ghosh, N Anand, and SM Sadaf, "**Enhanced Photon Extraction from GaN Point Defect Single-Photon Emitters via Top-Down Fabricated Nanowires,**" Photonics North 2026, Quebec, Canada (**Accepted**).
- S Adigbo, MA Rahman, MM Islam, DK Ghosh, **Md Fardin Ahmed**, C. Jenson, N. Anand, and SM Sadaf, "**Co₃O₄-Modified GaN Nanowires for High-Responsivity PEC Photodetection in Underwater Optical Communication,**" Photonics North 2026, Quebec, Canada (**Accepted**).

SKILLS

VLSI Design & EDA Tools: Cadence Suite (Layout, Schematic, Innovus, Genus, Xcelium), TSMC 16nm FinFET, DRC/LVS, RTL-to-GDSII, Verilog, PSPICE, Simulink, Quartus II

Instruments: Spectrum analyzers, Oscilloscopes, Signal generators, HBT Interferometry, APD, Nanopositioners, Soldering, Laser, Source Meter, DAQ, TimeTagger, KeySight Device Analyzer

Modeling & Simulation: CST, Lumerical FDTD, Material Studio

Programming: Python (Scripting, Automation, DAQ), MATLAB, C, OOP, LabVIEW

AI: Advanced ML/DL, TensorFlow, PyTorch, sklearn

Development Tools and Visualization: VScode, PyCharm, Jupyter, Origin, Seaborn, Matplotlib

UNDERGRADUATE PROJECTS [\[link\]](#)

- Design of an SPI Interface (2024)
- Experimental Setup to Measure the Radiation Pattern of a LED (2024)
- Low-Power Bandgap Reference Design in SKY130 (2024)
- Design and Verification of an 8-Bit ALU from RTL to GDSII (2024)
- Advanced ML Techniques for GHI Forecasting (2021)
- Digital Cash Register (2022)
- 1:8 Analog De-multiplexer with MOSFETs (2023)
- F450 Quadcopter with Ardupilot APM 2.8 (2019)
- Voice-Controlled Obstacle Avoiding Robot using Arduino and HC-05 (2019)
- Gesture-Controlled Robotic Navigation System (2022)

TEACHING AND MENTORING

Mentored and trained 10+ junior researchers in experiments, simulation, coding, and laboratory measurement techniques, strengthening their technical and scientific skills. Provided university level tutoring in Electronics and Solid State Physics to students from UBC and Monash University. Tutored and guided 70+ students in Bangladesh for competitive university admission examinations, with extensive instruction in Physics, Mathematics, and Chemistry. Brings clarity in explaining complex ideas, enthusiasm for student development, and a strong commitment to academic mentorship.

LEADERSHIP AND PROFESSIONAL ENGAGEMENT

- **Administrator, Student Association (CEISME), INRS-EMT** 2025
Led student welfare initiatives and campus events within the graduate research environment.
- **Member, IEEE Photonics Society, INRS-EMT** 2025
Supported invited seminars and expanded research networking in photonics.
- **President, IEEE Electron Devices Society Student Branch, BUET** 2023-2024
Organized and led seminars, invited talks, competitions, outreach, and 3MT Thesis programs. Managed to secure IEEE funding for BUET which substantially expanded student engagement.
- **General Secretary, BUET Optical and Photonics Society** 2023-2024
Organized workshops (Lumerical, MEEP), and academic events to promote research exposure.
- **Lead Organizer, EEE Day, BUET** 2023-2024
Participated in organizing major technical, academic, and cultural events for EEE Day in 2023 & 2024.
- **Member, Badhan, Bangladesh University of Engineering and Technology** 2021
Coordinated emergency blood donation logistics and community volunteer service.

INTERNATIONAL COLLABORATION

- **University of Technology Sydney (UTS), Australia**
Collaborating with **Prof. Igor Aharonovich** on GaN single photon emitters, with emphasis on growth dependent emission behaviour, defect understanding, and quantum device applications.
- **Okayama University, Japan**
Collaborating with **Prof. Masazumi Fujiwara** on GaN based quantum biosensing, including quantum defect characterization and ODMR related measurements for future sensing applications.

RELEVANT COURSEWORK

Theory Courses: Analog Electronics, VLSI, Analog Integrated Circuit and Design, Nonlinear Optics, Solid State Devices, Compound Semiconductor Devices, Electronic Properties of Materials, Semiconductor and Nano Devices, Optoelectronics, Electromagnetics, Digital Electronics, Microprocessor and Embedded System, Digital Signal Processing, Control System, Power Electronics, Energy Conversion, Electrical Circuits, Programming Language in C & C++.

Laboratory Courses: VLSI Circuit & Design, Microprocessor & Embedded System, Analog Integrated Circuits Lab, Control System Lab, Optoelectronics Lab, Digital Electronics Lab, Energy Conversion Lab, Electronic Circuits Lab, C & C++ Lab

CERTIFICATES

- **Deep Learning Specialization** – Andrew Ng, Stanford University (2020)
- **Machine Learning Specialization** – Andrew Ng, Stanford University (2020)
- **Python for Everybody Specialization** – Charles Severance, University of Michigan (2020)